

OCULUS | Keratograph 5M



DESCRIPTION OF DATA INTERFACES

System for measuring and analysing the cornea topography

Notes on this instruction manual

The described procedures are supplements to the Keratograph 5M. To ensure safe operation, it is essential that you use the device correctly. For this reason you should familiarize yourself thoroughly with the contents of this instruction manual before operating the device. In particular, pay attention to the safety instructions.

Read the instructions and user manual of the Keratograph 5M. There you will find all the information for using the device.

Due to ongoing development, the diagrams shown may depict minor changes to the actual device delivered.

Depending on your operating system, the images shown below may differ from what you see on your PC.

If you have any queries or would like additional information about your device, do not hesitate to call or send us a mail or a fax. Our service team will gladly assist.

OCULUS Optikgeräte GmbH



OCULUS is certified according to DIN EN ISO 13485, setting high standards of quality where development, manufacture, quality assurance and service regarding the entire range of products are concerned..

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1 Safety instructions

The described procedures are supplements to the Keratograph 5M. The safety instructions of the instruction manual apply.

- ➔ Read these instructions carefully.
- ➔ Read the instructions for the equipment employed.

Graphic symbols used



Note

Identifies situations which can result in erroneous results, or denotes application information as well as useful or important information.

2 Information on the Data Interfaces

This manual describes the settings and data formats of the data interfaces to export data of the Keratograph 5M to doctors' practice software or the hospital software.

The following procedures are described:

- Activate the data interfaces for the export, [sec. 3, page 2](#)
- Data interfaces for cornea data and contact lens data, [sec. 4, page 3](#)

Depending on the operating system, the screens may vary.



Note

OCULUS is not liable for damage caused by a faulty installation.

- ➔ Follow these instructions to install the software.

3 Activate the data interfaces for the export

- ➔ Start the Keratograph 5M software.
- ➔ Select a patient.
- ➔ Select an examination.
- ➔ Select Settings > Miscellaneous Settings.
- ➔ Select the tab [Export].

This screen is displayed:

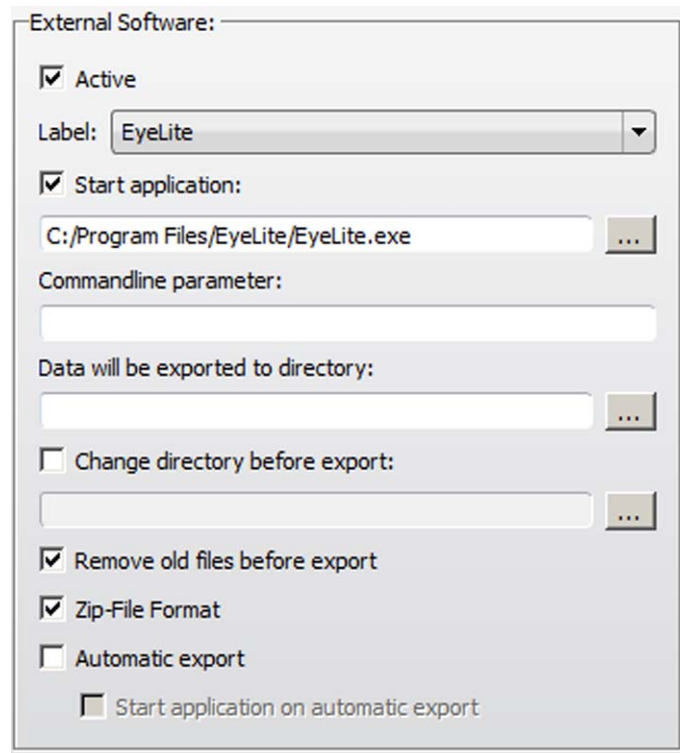


Fig. 3-1: Settings „External Software“

- **Label:** Activate and enter the name of the external software.
- **Start application:** Activate and enter the name of the .exe file of the external software.
- **Command line parameter:** Enter additional parameter required by the external software
- **Data will be exported to directory:** Transfer folder of the data interface files
- **Change Directory before Export:** Activate and enter the directory to execute the external software. If its function ist not activated, the programm folder „Topo“ will be set.
- **Remove old files before export:** Old files will be deleted
- **Zip-File Format:** Files will be zipped and exported
- **Automatic Export:** Implement the automatic export for external software. Enable [Start application on automatic export] checkbox for each external software.

4 Data Interfaces for Cornea Data and Contact Lens Data

The measured values of the cornea and the chosen contact lens can be returned by means of this interface to the patient data management.

Therefore the following files in the directory are created when the examination program is closed:

PATIENT . TXT	<i>sec. 4.1, page 4</i>
EXAM . TXT	<i>sec. 4.2, page 4</i>
EY - PARA . DAT	<i>sec. 4.3, page 5</i>
CORNEA . OS (In case an examination of the left eye was taken or loaded)	CORNEA . OD (In case an examination of the right eye was taken or loaded) <i>sec. 4.4, page 6</i>
CORNEA_F . OS (In case an examination of the left eye was taken or loaded)	CORNEA_F . OD (In case an examination of the right eye was taken or loaded) <i>sec. 4.5, page 7</i>
CURVAT . OS (In case an examination of the left eye was taken or loaded)	CURVAT . OD (In case an examination of the right eye was taken or loaded) <i>sec. 4.6, page 7</i>
PUPIL . OS (In case an examination of the left eye was taken or loaded)	PUPIL . OD (In case an examination of the right eye was taken or loaded) <i>sec. 4.7, page 7</i>
CENTER . OS (In case an examination of the left eye was taken or loaded)	CENTER . OD (In case an examination of the right eye was taken or loaded) <i>sec. 4.8, page 8</i>
CL - PARA . DAT	<i>sec. 4.9, page 8</i>
RF - PARA . DAT	<i>sec. 4.10, page 9</i>
EY - EXZE . DAT	<i>sec. 4.11, page 10</i>
PIC - OS . BMP (In case an examination of the left eye was taken or loaded)	PIC - OS . BMP (In case an examination of the right eye was taken or loaded) <i>sec. 4.12, page 11</i>

CURVAT_F.OD/OS with extrapolated curvature data are also exported.

These files contain the data of the latest examination as well as of the latest simulated contact lens of the right or the left eye.

4.1 Patient Data: PATIENT.TXT

This file contains the patient's data.

Example

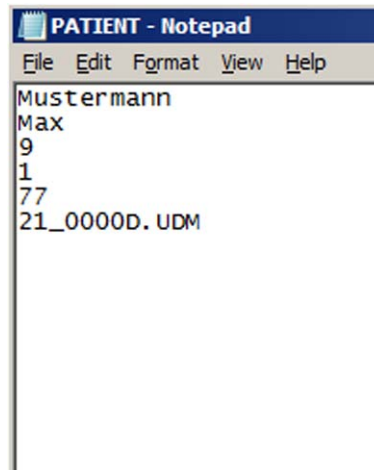


Fig. 4-1: Patient's data

Line 1: last name

Line 2: first name

Date of birth

Line 3: day

Line 4: month

Line 5: year

Line 6: patient file name in database

4.2 Examination Data: EXAM.TXT

This file contains the data of the examination.

Example

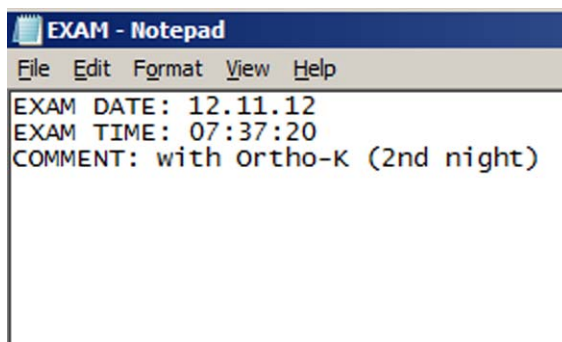


Fig. 4-2: Data of the examination

Line 1: date of examination

Line 2: time of examination

Line 3: comment to the examination

4.3 Keratometer Data: EY-PARA.DAT

In this file the measured values of the cornea of the right or left eye are stored according to the following scheme:

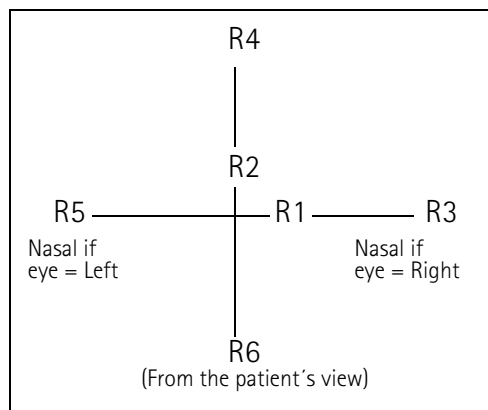


Abb. 4-3: Scheme of the keratometer data

First of all, the eye is named: (OS/OD)

Then the central curvatures as well as the sagittal radii in 30° are stored in the following order:

- R1. Horizontal central radius
- R2. Vertical central radius
- R3. Horizontal sagittal radius in 30° (3 o'clock)
- R4. Vertical sagittal radius in 30° (12 o'clock)
- R5. Horizontal sagittal radius in 30° (9 o'clock)
- R6. Vertical sagittal radius in 30° (6 o'clock)

Finally, the axis of the astigmatismus (flat meridian) is stored.

OS is always on the top, OD on the bottom. In front of OD is a separating line: „- - - - -“

If there was no measurement taken for an eye, only empty lines are shown:

Example 1

OS
7.86
7.75
7.84
7.80
8.13
7.89
21.2
- - - - -
OD
7.80
7.40
8.34
7.58
7.92
7.62
5.0

Example 2

OS

- - - - -
OD
7.80
7.40
8.34
7.58
7.92
7.62
5.0

Example 3

OS
7.86
7.75
7.84
7.80
8.13
7.89
21.2
- - - - -
OD

4.4 Cornea Data Files: CORNEA.OS and CORNEA.OD

If an examination has been taken or loaded with the Keratograph software, the software creates a file in which the cornea surface is stored by means of zone height values. Depending on the selected eye the file CORNEA.OD (right) or CORNEA.OS (left) is created.

- If no examination has been used, no cornea data file is stored.
- If only an examination of the right (left) eye has been taken or loaded, only the file CORNEA.OD (CORNEA.OS) is stored.
- If examinations of both eyes have been taken or loaded, both files CORNEA.OD and CORNEA.OS are stored.
- If two examinations of one eye have been used, only the data of the last examination used are stored.

The cornea data are stored by using zone height values of half meridians. 100 half meridians are stored in steps of 0.1mm from the center to the periphery.

Example

```

Seg: 0  y= 0.00  x= 0.000000000000
Seg: 0  y= 0.10  x= 0.000667879663
Seg: 0  y= 0.20  x= 0.002695877245
Seg: 0  y= 0.30  x= 0.006080514826
Seg: 0  y= 0.40  x= 0.010817467380
Seg: 0  y= 0.50  x= 0.016903643213

```

Seg: angle of the half meridian

Y: distance to the center

X: zone height

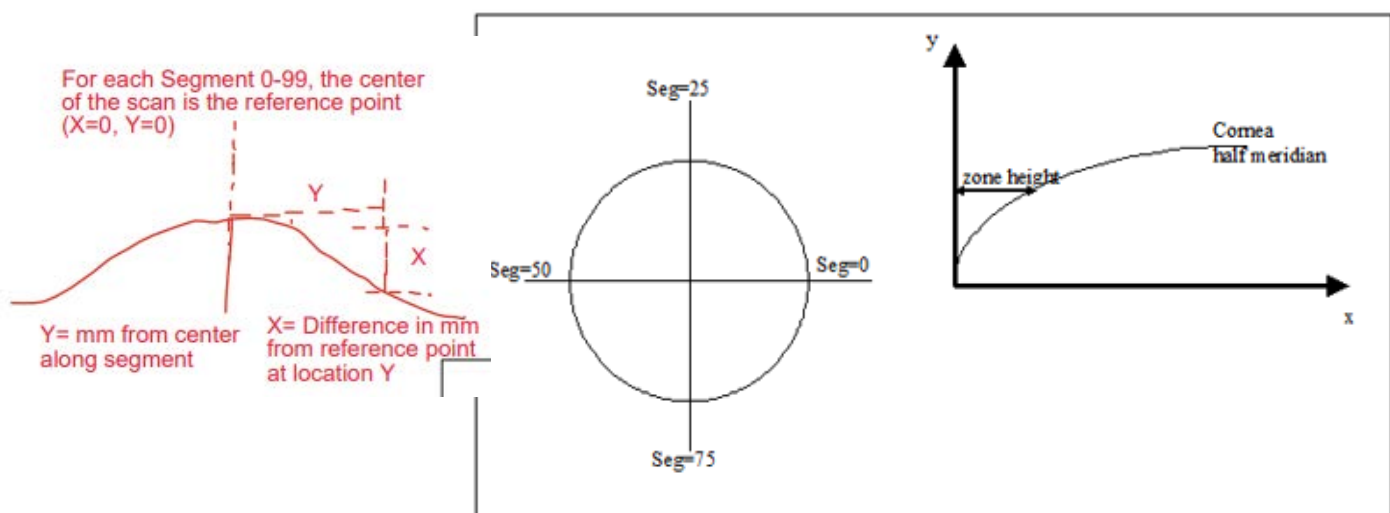


Fig. 4-4: Cornea data

4.5 Cornea Data Files: CORNEA_F.OS and CORNEA_F.OD

This file is almost identical to CORNEA.OS and CORNEA.OD, [sec. 4.4, page 6](#), but including the interpolated areas.

4.6 Cornea Data Files: CURVAT.OS and CURVAT.OD

This file is almost identical to CORNEA.OS and CORNEA.OD, [sec. 4.4, page 6](#). But instead of zone height it shows the curvature value at each location.

r(Sag): axial/sagittal curvature

r(Tan): tangential curvature

Example

Seg: 0	y= 0.00	r (Sag)=	8.77	r (Tan)=	8.89
Seg: 0	y= 0.10	r (Sag)=	8.77	r (Tan)=	8.87
Seg: 0	y= 0.20	r (Sag)=	8.79	r (Tan)=	8.86
Seg: 0	y= 0.30	r (Sag)=	8.80	r (Tan)=	8.85
Seg: 0	y= 0.40	r (Sag)=	8.81	r (Tan)=	8.84
Seg: 0	y= 0.50	r (Sag)=	8.82	r (Tan)=	8.84
Seg: 0	y= 0.60	r (Sag)=	8.83	r (Tan)=	8.82
Seg: 0	y= 0.70	r (Sag)=	8.83	r (Tan)=	8.79
Seg: 0	y= 0.80	r (Sag)=	8.83	r (Tan)=	8.77
Seg: 0	y= 0.90	r (Sag)=	8.82	r (Tan)=	8.75
Seg: 0	y= 1.00	r (Sag)=	8.81	r (Tan)=	8.72

4.7 Pupil Margin Files: PUPIL.OS and PUPIL.OD

In this file the distance of the pupil margin to the center is given for 100 half meridians. The order corresponds to the cornea data.

Example

Seg: 0	y= 1.89
Seg: 1	y= 1.91
Seg: 2	y= 1.94
Seg: 3	y= 1.96
Seg: 4	y= 1.99
Seg: 5	y= 2.02
Seg: 6	y= 2.06
Seg: 7	y= 2.10
Seg: 8	y= 2.13
Seg: 9	y= 2.18
Seg:10	y= 2.22
Seg:11	y= 2.27
Seg:12	y= 2.31
Seg:13	y= 2.36
Seg:14	y= 2.41

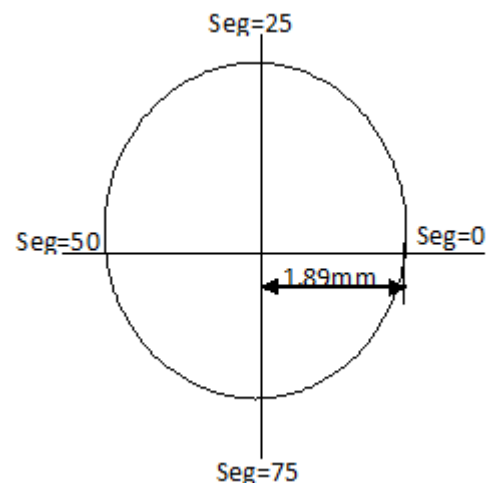


Fig. 4-5: Pupil margin files

4.8 Center Files: CENTER.OS and CENTER.OD

In this file the position of the center of the pupil, the center of the iris, the diameter of the pupil and the horizontal diameter of the iris are given.

Example

```

Iris Center      : x=-0.34
Iris Center      : y=-0.26
Iris Diameter    : 11.45
Pupil Center     : x=-0.49
Pupil Center     : y=0.49
Pupil Diameter   : 4.85
  
```

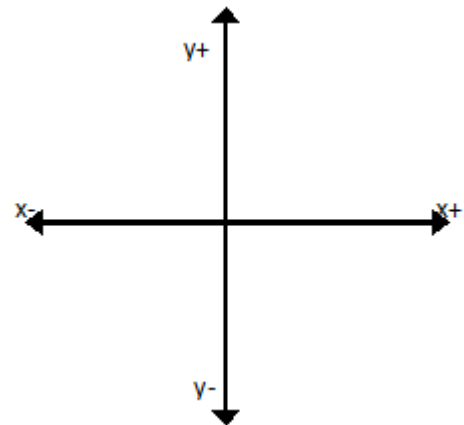


Fig. 4-6: Center files

4.9 Contact Lens Data: CL-PARA.DAT

In this file the most important data of the contact lens back surfaces are stored:

Manufacturer, lens type, central radii, eccentricity, diameter.

These data are stored one after the other for both eyes OS and OD. .

Example 1

OS

OD

```

Lens: Wöhlk
Typ: conflex toric
r=8.00/7.60mm
Ecc=0.00
D=9.40mm
  
```

Example 2

OS

```

Lens: Müller-Welt
Typ: CNM Innen toric
r=8.00/7.60mm
Ecc=0.00
D=9.00mm
  
```

OD

```

Lens: Wöhlk
Typ: conflex-air
r=7.80mm
Ecc=0.40
D=9.80mm
  
```

Example 3

OS

```

Lens: Hecht
Typ: AS6
r=7.75mm
Ecc=0.60
D=9.60mm
  
```

OD

Every parameter is stored in another line.

The texts „Lens:“, „Typ:“, „r=“, „Ecc“ and „D=“ are always the same.

If there was no lens simulated for an eye, so only empty lines are stored for this eye.

4.10 Refraction Data: RF-PARA.DAT

In this file the refraction of the glasses, the type of the contact lens and refractive power of the contact lens are stored:

These data are stored one after the other for both eyes OS and OD.

Example 1

OS

Eye Sph: 1.50
 Eye Cyl: -0.50
 Eye Axs: 0
 Eye cvd: 0
 Lens : Rotation symmetric
 Sph : 0.75
 Cyl :
 Mat :

OD

Eye Sph: 3.75
 Eye Cyl: -2.00
 Eye Axs: 90
 Eye cvd: 15
 Lens : Back toric
 Sph : 1.00
 Cyl :
 Mat : Zeiss A90

Example 2

OS

OD

Eye Sph: 3.75
 Eye Cyl: -2.00
 Eye Axs: 90
 Eye cvd: 15
 Lens : Soft
 Sph : 4.00
 Cyl : -2.25
 Mat :

Every parameter is stored in another line.

The texts "Eye Sph:", "Eye Cyl:",... are always the same.

If there was no lens simulated for an eye, so only empty lines are stored for this eye.

4.11 Eccentricities: EY-EXZE.DAT

In this file the eccentricities of the right or left eye are stored according to the following scheme:

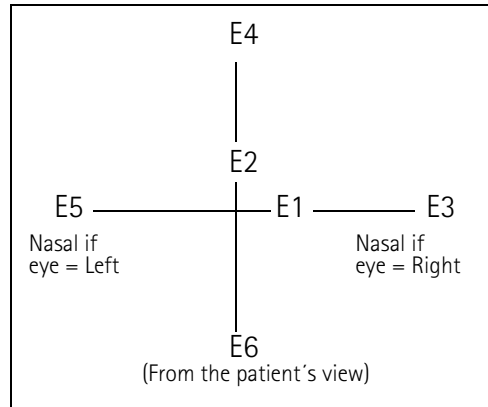


Fig. 4-7: Scheme „Eccentricities“

First of all, the eye is named: (OS/OD)

Then the eccentricities (in 30°) are stored in the following order:

- E1. Mean eccentricity horizontal (from E3,E5)
- E2. Mean eccentricity vertical (from E4,E6)
- E3. Eccentricity horizontal 30° (3 o'clock)
- E4. Eccentricity vertical 30° (12 o'clock)
- E5. Eccentricity horizontal 30° (9 o'clock)
- E6. Eccentricity vertical 30° (6 o'clock)
- E7. Mean eccentricity (from E3,E4,E5,E6)

OS is always on the top, OD on the bottom. In front of OD is a separating line: „- - - - -“

If there was no measurement taken for an eye, only empty lines are shown:

Example 1

OS
0.53
0.69
0.40
0.62
0.65
0.76
0.61
 - - - - -
OD
0.51
0.63
0.78
0.53
0.24
0.73
0.57

Example 2

OS

 - - - - -
OD
0.51
0.63
0.78
0.53
0.24
0.73
0.57

Example 3

OS
0.53
0.69
0.40
0.62
0.65
0.76
0.61
 - - - - -
OD

4.12 Image of eye: PIC-OS.BMP and PIC-OD.BMP

This file contains an image of the examined eye.

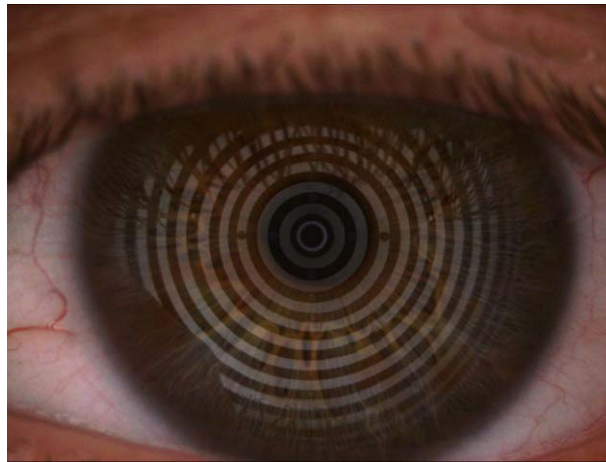


Fig. 4-8: Example of image:PIC-OS

5 Address of the Manufacturer and Service Department

Our service department or authorised representatives will furnish you with additional information.

Address of the manufacturer and service department:

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